

Potential Threats and Mitigation for a Small Office Network

1. Unauthorized Access

Threat:

Unauthorized users gaining access to network resources.

Mitigation:

- Implement strong authentication (e.g., WPA3 for Wi-Fi, multi-factor authentication for critical systems).
- Enforce role-based access control (RBAC).
- Regularly review and update access control policies.

2. Malware and Ransomware

Threat:

Malicious software infecting devices, leading to data loss or ransom demands. **Mitigation:**

- Deploy endpoint protection (antivirus, anti-malware solutions).
- Regularly update software and firmware.
- Educate employees on phishing and malicious attachments.

3. Insider Threats

Threat: Employees or guests misusing access, intentionally or unintentionally. **Mitigation:**

- Implement user activity monitoring.
- Implement access list protocol
- Enforce least privilege principles.
- Conduct cybersecurity awareness training.

4. Wireless Network Attacks

Threat:

Rogue access points, deauthentication attacks, and eavesdropping.

Mitigation:

- Use strong encryption (WPA3) and disable WPS.
- Regularly scan for rogue access points.

5. IoT Device Vulnerabilities

Threat:

IoT devices being exploited as attack vectors.

Mitigation:

- Segment IoT devices into a separate VLAN.
- Change default credentials and update firmware regularly.
- Disable unnecessary services and ports.

6. Network Eavesdropping (MITM Attacks)

Threat:

Attackers intercepting communications between devices. **Mitigation:**

- Use encrypted protocols (HTTPS, SSH, VPNs).
- Enable network segmentation and VLANs.
- Deploy intrusion detection/prevention systems (IDS/IPS).

7. Denial-of-Service (DoS) Attacks

Threat:

Attackers overwhelming network resources, causing downtime.

Mitigation:

- Implement rate limiting and traffic filtering on routers.
- Deploy firewalls with DoS protection.
- Monitor network traffic for anomalies.

8. Phishing and Social Engineering

Threat:

Attackers tricking employees into revealing sensitive information.

Mitigation:

- Conduct regular security awareness training.
- Implement email filtering and anti-phishing solutions.
- Enforce strict verification procedures for sensitive requests.

9. Physical Security Breaches

Threat:

Unauthorized access to network hardware.

Mitigation:

- Lock network devices in secure locations.
- Use surveillance and access control systems.
- Restrict USB and external device usage on company systems.
- Disable unnecessary ports

10. Misconfigured Network Devices

Threat

: Poorly configured routers, switches, and access points leading to vulnerabilities. **Mitigation:**

- Regularly audit and review configurations.
- Disable unused services and default accounts.